

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

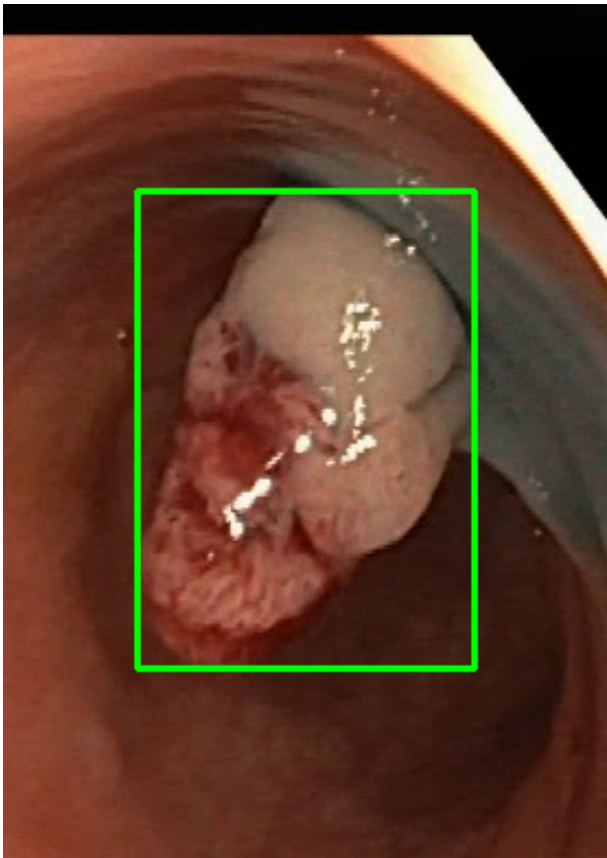
Video ID:	496f3fe6-b069-47aa-80d0-cfdda1d28fc0
Total Frames:	3510
Frame Rate:	25.0 fps (assumed)
Duration:	140.4 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	1658	✓
RT-DETR Detections	1975	✓
MedSAM2 Detections	1878	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	Count	Percentage
HIGH RISK	3	100.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 424 frames)



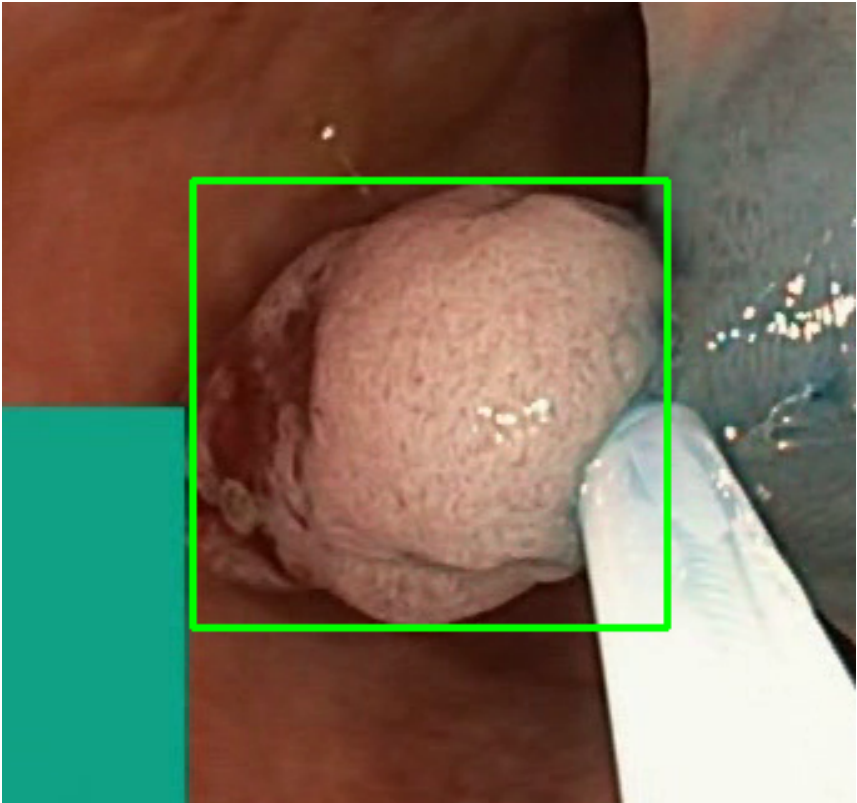
Feature	Value	Interpretation
Frame Index	2003	Frame 2003 of 3510
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 Unavailable for this video)
POLYP TYPE	MALIGNANT_POLYP	Type confidence: 92.0%
Redness Score	0.230	Color-based vascularization
Relative Radius	24.2%	Polyp size as % of frame diagonal
Texture Score	0.195	Surface morphology
Vessel Visibility	0.547	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	MALIGNANT_P OLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	90.9%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #2 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 1005 frames)



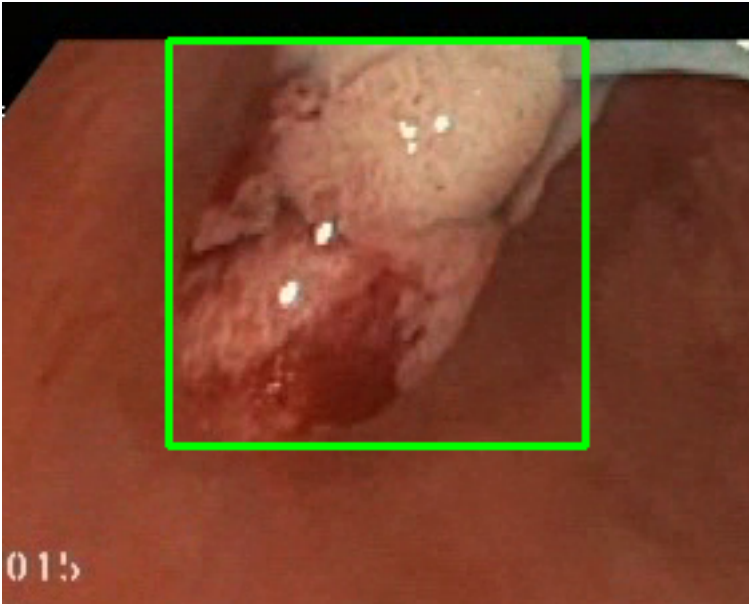
Feature	Value	Interpretation
Frame Index	2956	Frame 2956 of 3510
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_PO LYP	Type confidence: 92.0%

Redness Score	0.238	Color-based vascularization
Relative Radius	20.2%	Polyp size as % of frame diagonal
Texture Score	0.180	Surface morphology
Vessel Visibility	0.812	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	MALIGNANT_P OLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	94.1%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #3 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 30 frames)



Feature	Value	Interpretation
Frame Index	1678	Frame 1678 of 3510
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)

POLYP TYPE	BLEEDING_PO LYP	Type confidence: 92.0%
Redness Score	0.255	Color-based vascularization
Relative Radius	20.8%	Polyp size as % of frame diagonal
Texture Score	0.146	Surface morphology
Vessel Visibility	0.700	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	MALIGNANT_P OLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	91.6%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:

A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 3 (100.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.